

Fengzhou Wang

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Educational Background

- Jilin University (985), School of Mechanical and Aerospace Engineering, Industrial Engineering (National First-Class Discipline), Overall Ranking: 1/60, Sep 2021 - Jun 2025
- Zhejiang University (985), National Institute of Outstanding Engineers, Industrial Engineering and Management (Master), Sep 2025 - Jun 2028
- Core Courses: Operations Research, Production Planning and Control, Human Factors Engineering, System Modeling and Simulation, Management Information Systems, Logistics Engineering, Python Data Analysis, Supply Chain Management, Engineering Management Mathematics, Big Data Analysis and Mining, Operations Management, etc.

Professional Skills

- Proficient in MS Office: Skilled in data processing with EXCEL, PPT creation and drawing with PowerPoint, and drafting various documents with Word.
- Proficient in Visio/Xmind: Capable of process visualization design, project structured thinking organization and requirement breakdown.
- Proficient in IE thinking and methods: Skilled in analyzing and planning projects with characterization tools such as Gantt charts and fishbone diagrams; good at proposing problem solutions with the TRIZ method; familiar with logistics facility planning methods such as SLP, relevant optimization theories of operations research and basic project management methods.
- Familiar with Python: Proficient in Pycharm operation, able to conduct AI-assisted programming with Trae and Cursor; skilled in data analysis with common libraries such as NumPy, Pandas and Matplotlib; have a basic understanding of the use of deep learning frameworks such as TensorFlow and PyTorch.
- Master UI and product design: Proficient in Axure, PS and Wordpress, capable of basic visual interface design and interaction design.

Research Experience

- Segmented parabolic adjustment of the FAST reflector utilizing spatial coordinate rotation transformation, Measurement Science and Technology, Jun 2024 (Accepted, First Author, CAS Tier 3, IF=2.7): Proposed a segmented parabolic adjustment method for the FAST reflector based on spatial coordinate rotation transformation. Constructed a segmented ideal parabolic model through coarse and fine grid search and BP algorithm optimization, improving

the reflection efficiency from 77.92% to 95.56%, and providing a new strategy for the high-efficiency focusing of the reflector of large-aperture radio telescopes.

- Dialogues between Adam and Eve: Exploration of Unknown Civilization Language by LLM, ICLR 2025 Workshop World Models, Mar 2025 (Accepted, Co-First Author): Constructed the "Adam-Eve" multi-agent dialogue system, where Adam generates random languages with rules of different difficulty levels, and Eve simulates the language acquisition process of unknown civilizations through strategies such as reinforcement learning. Revealed the language emergence mechanism of LLM without prior knowledge, and verified that reinforcement learning significantly improves language learning efficiency and semantic similarity and other indicators.
- A novel Failure Mode and Effect Analysis considering uncertain evaluations under rule-based Bayesian network modelling, Chinese Journal of Mechanical Engineering, Nov 2025 (Required Reviews Complete, Co-First Author, CAS Tier 3, IF=4.7): Developed an improved FMEA method integrating Evidential Reasoning (ER) and Rule-based Bayesian Network (RBN). Solved the problems of uncertain expert evaluation and inaccurate RPN calculation in traditional FMEA through MDM-BWM weight fusion and RBN risk propagation analysis, and significantly improved the risk quantification accuracy of CNC lathe reliability analysis.
- BiLSTM-Based Aircraft Engine RUL Prediction Method Integrated with Attention Mechanism, 2025 7th International Conference on System Reliability and Safety Engineering (EI, Accepted, Co-First Author & Corresponding Author): Proposed a WT-Attention-BiLSTM hybrid model to solve the problems of high noise and long-range dependence in aircraft engine RUL prediction. Used wavelet transform (Db4 basis) for signal denoising and data augmentation, combined BiLSTM to extract bidirectional temporal features, and introduced the Attention mechanism to focus on key degradation stages. On the NASA C-MAPSS dataset, the comprehensive results are significantly better than traditional CNN and SVR models.
- Research on Motion Optimization Method Based on Digital Twin and Reinforcement Learning, Undergraduate Graduation Thesis, Apr 2025: Proposed a collaborative optimization framework of improved DDPG and other algorithms combined with sports science, through a virtual-real closed-loop verification mechanism and multi-objective constraint design, and verified in tennis movements. Provided an extensible algorithm paradigm for interdisciplinary motion optimization, designed a basic system interface and connected it to Deepseek for auxiliary decision-making.
- TADLE: A Temporal Attention-based Dual-Stream LSTM Heterogeneous Ensemble for Electricity Price Trend, Electric Power Systems Research, Mar 2026 (Submitted, Corresponding Author, CAS Tier 3, IF=4.2): Proposed a hybrid ensemble model integrating LightGBM and dual-stream LSTM to optimize the cost sensitivity of electricity market price

prediction. Achieved the optimal results of $F1=0.8545$, $AUC=0.9477$ and 1.6% reduction in CostScore on real datasets, providing a solution with both accuracy and economy for power grid transactions.

- DO-XGB: A Badminton Prediction Model for Transceiver Robot, ROBOTICS AND AUTONOMOUS SYSTEMS (Submitted, First Author, CAS Tier 2, IF=5.2): Proposed a DO-XGB trajectory prediction algorithm integrating physical priors and multi-task learning to solve the perception bottleneck of badminton robots in data-scarce scenarios. Innovatively designed a physics-inspired simulation data generation method and developed a regression-classification joint learning architecture to effectively capture the dynamic characteristics of trajectories. Experiments show that while maintaining 100% physical boundary prediction accuracy, the model reduces the 3D landing point prediction error to 0.1335m with an F1 Score of 0.9805, which is significantly better than baseline models such as Transformer.
- An Agricultural E-commerce Platform and Its Purchasing and Selling Method, Patent No. 202310046469.8[P], 23 Jun 2023 (In Substantive Examination, Invention Patent)
- A Dual-Wheel Transceiving Integrated Tennis Training Robot and Its Operation Method, Patent No. 202410206515.0[P], 03 May 2024 (In Substantive Examination, Invention Patent)
- An Intelligent Road Snow and Ice Removal Robot, Patent No. 202410406669.4[P], 07 Apr 2024 (Preliminarily Examined and Qualified, Invention Patent)
- A Transceiving Integrated Badminton Training Intelligent Robot, Patent No. 202410712122.7[P], 20 Aug 2024 (In Substantive Examination, Invention Patent)
- An AI-Based Creation Guidance System for Graphic Art Works, Patent No. 202410355040.1[P], 27 Mar 2024 (Preliminarily Examined and Qualified, Invention Patent)
- A Traction-Type Rotor Rake, Patent No. 202420671955.9[P], 22 Nov 2024 (Authorized, Utility Model Patent & Invention Patent)
- A Nanosensor for Detecting MMP9 Markers and Its Application, Patent No. 202410186726.2[P], 20 Feb 2024 (Preliminarily Examined and Qualified, Invention Patent)
- A Structure and Manufacturing Method of a Magnetic Soft Robot, Patent No. 202410524320.0[P], 02 Jul 2024 (In Substantive Examination, Invention Patent)
- A Path Planning Method for Intelligent Tractors, Patent No. 202410399235.6[P], 03 Apr 2024 (In Substantive Examination, Invention Patent)

Project Experience

National College Students' Innovation and Entrepreneurship Training Program (Participant, Completed), May 2022 - May 2023

- Project Content: Designed and manufactured a high-performance fire rescue robot equipped with a high obstacle-surmounting metal crawler chassis and a seat-type high-strength cockpit,

adopting a high-temperature resistant protective material interlayer. Realized remote rescue by identifying fire and trapped persons based on the YOLOv5 model with an infrared camera.

Provincial College Students' Innovation and Entrepreneurship Training Program (Project Leader, Completed), May 2023 - May 2024

- Project Content: Designed an intelligent multi-functional road snow and ice removal robot including a driving module, a snow blowing module, an ice drilling module, a bucket module, a spray module and a vision module. Realized multi-functional operation through ice and snow road identification and path planning based on FCN with a camera.
- Achievements: One invention patent (Preliminarily Examined and Qualified, First Inventor, Application No.: 202410406669.4)

National College Students' Innovation and Entrepreneurship Training Program (Participant, Completed), May 2023 - May 2024

- Project Content: Aiming at the possible impact of generative AI on graphic art creation, designed an AI-based creation guidance system for graphic art works from the perspectives of creation ethics and originality, which can conduct evaluation and guidance based on submitted works.
- Achievements: One invention patent (Preliminarily Examined and Qualified, Second Inventor, Application No.: 202410355040.1), one Chinese paper

Competition Experience

- Intelligent Equipment Management Practice, National Third Prize of Industrial Engineering and Lean Management Innovation Competition, Mechanical Engineering Innovation and Creativity Competition (2/4), Nov 2022
Project Content: Aiming at the pain points of equipment management in a small and medium-sized enterprise, developed an intelligent management platform. Realized early identification of equipment failures and energy consumption optimization by applying equipment classification rules, ARIMA and SVR state prediction models and TnPM management concepts, effectively improving equipment OEE and reducing maintenance costs.
- Enterprise Digitalization for Information Flow Optimization, National Second Prize of Tsinghua University IE Bright Sword Industrial Engineering Application Case Competition (1/5), Oct 2023
Project Content: Optimized the enterprise production process and warehouse management system through Plant Simulation, RFID technology and information field theory, designed a

WMS/MES information system and introduced the Actor-Critic algorithm to improve logistics scheduling, significantly enhancing material turnover efficiency and production transparency.

- Research and Implementation of Equipment Management System, National Second Prize of Industrial Engineering and Lean Management Innovation Competition, Mechanical Engineering Innovation and Creativity Competition (1/5), Oct 2023

Project Content: Integrated functions such as condition monitoring, fault prediction (Barlow model / adaptive 3sigma rule) and maintenance into one system. Realized standardization, leanization and informatization of equipment management by combining industrial engineering methods and intelligent algorithms, helping enterprises improve production efficiency and equipment reliability.

- Optimal Design of Human-Robot Collaborative Intelligent Production Line, National Second Prize of the Preliminary Round, "Siemens Cup" China Intelligent Manufacturing Challenge (1/3), Jul 2023

Project Content: Aiming at factories with multi-variety and small-batch production, optimized the production line layout and operation task allocation based on human factors engineering and Plant Simulation, introduced collaborative robots to improve production flexibility, and ensured production line reliability through the HEART method, effectively reducing labor load and improving production efficiency.

- Ginseng and Cordyceps Integrated Oral Microemulsion, National Second Prize of National College Students' Life Science Competition (Innovation and Entrepreneurship Group) (2/7), Aug 2023

Project Content: Designed and developed a health care oral microemulsion product integrated with ginseng and cordyceps, completed product process and packaging design, business model construction, etc., and provided an immune health solution for frail people.

- AI-Based Plant Disease and Pest Early Warning System for Plant Protection, National Third Prize of National College Students' Energy Conservation and Emission Reduction Social Practice and Technology Competition (3/7), Aug 2023

Project Content: Designed and developed a set of AI-based device for monitoring, early warning and evaluation of crop diseases and insect pests, powered by solar energy. Established a target detection neural network based on the YOLOv7 model to identify corresponding diseases and insect pests and issue early warnings, and carried out pest trapping and killing, yield evaluation, etc.

- Biomimetic Principle-Based Lithium Battery Flame-Retardant Microcapsules, National Bronze Prize of China International College Students' Innovation and Entrepreneurship Competition (Champion of Jilin Province, 5/16), Apr 2024

Project Content: Participated in the design and development of a lithium battery flame-

retardant microcapsule based on the biomimetic principle of immune cells, and the relevant results were published in a Cell sub-journal. The team is committed to improving battery safety and promoting the transformation and industrialization of technological achievements.

- Application of Early Screening Methods for Colorectal Cancer, National Third Prize of National College Students' Life Science Competition (Innovation and Entrepreneurship Group) (4/7), Aug 2024

Project Content: Participated in the design and development of an MMP9 marker detection technology and supporting kit based on nanosensors, designed a mini-program interactive interface, and promoted the convenience and accuracy of early screening technology for colorectal cancer.

- Scheme Design and Research of Valve Body Casting System, National Second Prize of Casting Process Design Competition, Mechanical Engineering Innovation and Creativity Competition, Aug 2024

Project Content: Conducted 3D modeling, drew CAD drawings and drafted process specifications according to the competition requirements, proposed an optimized casting process scheme and applied AnyCasting software for simulation, and improved casting quality and production efficiency while reducing production costs through structural parameter design and process planning.

- A New Generation of High-Performance Fire Rescue Robot, Gold Prize of Jilin Province, 9th "Internet +" China International College Students' Innovation and Entrepreneurship Competition (2/3), Oct 2023

Project Content: Designed and manufactured a high-performance fire rescue robot equipped with a high obstacle-surmounting metal crawler chassis and a human factors engineering high-strength cockpit, adopting a high-temperature resistant protective material interlayer. Realized remote rescue by identifying fire and trapped persons based on the YOLOv5 model with an infrared camera.

- Human-Robot Collaborative Digital Tennis Training Equipment, Bronze Prize of Jilin Province, 9th "Internet +" China International College Students' Innovation and Entrepreneurship Competition (2/12), Oct 2023

Project Content: Designed and developed a dual-wheel transceiving integrated tennis confrontation training robot and wearable digital twin visual training equipment, dynamically evaluated the trainer's hitting ability and adaptively adjusted the training difficulty to realize personalized training, improving tennis training efficiency and reducing costs.

Campus Experience

Honors and Awards

- Scholarships and Grants: National Encouragement Scholarship *3, Individual Scholarship *1, China Automotive Technology and Research Center Scholarship *1, Integrity Grant *1, Bosen Electric Grant *1, etc.
- Honorary Titles: Excellent Student of the School *2, Excellent Student Cadre of the School *1, Excellent League Member *1, Outstanding Contribution Award for Epidemic Prevention and Control *1, Contribution Award for Cultural and Sports Activities *1
- Special Experience: Student Representative of the 28th and 29th Student Congress of Jilin University

Student Cadre Experience

- Class Position: Sports Committee (Oct 2021 - Mar 2023), League Branch Secretary & Dormitory Monitor (Mar 2023 - Jun 2025), Monitor (Sep 2025 - Present)
- Association Position: Minister of Publicity Department, China Volunteer Association (Oct 2022 - Oct 2023)
- School Position: Deputy Minister of Publicity Department, Communist Youth League Committee (Jun 2022 - Jun 2023), Minister (Jun 2023 - Jun 2024), Assistant to Freshman Counselor (Sep 2023 - Sep 2024), Assistant of the Office, Communist Youth League Committee (Oct 2022 - Oct 2024), etc.
- University Position: Person in Charge of Jiangxi Province, Admissions and Publicity Association of Jilin University (Dec 2022 - Dec 2023), Member of the Copywriting Group, Jilin University Youth Newspaper.

Social Practice and Volunteer Service

- Served as an epidemic prevention and control volunteer from Apr to May 2022, responsible for meal delivery and nucleic acid testing in the 11th Dormitory Building.
- Served as a volunteer for the 15th International Conference on Design and Manufacturing in Aug 2022, assisting in conference information registration and payment work.
- Winter Social Practice in Jan 2022: Campus Promotion (Person in Charge), organized campus publicity activities for more than 40 universities.
- Winter Social Practice in Jan 2023: Campus Promotion (Provincial Person in Charge of Jiangxi Province, Person in Charge of Jilin University Jiangxi Return Practice Group), won the University-Level Second Prize, Advanced Social Practice Team, Advanced Individual, and the

honor of "Excellent Volunteer" for 2023 Admissions and Publicity of Jilin University (the only one in the school).

- Summer Social Practice in Jul 2022: Community Service (Person in Charge of Youth Footprints Team), volunteer at the county library, drowning prevention publicity, epidemic prevention, etc.
- Summer Social Practice in Aug 2023: Rural Revitalization (Person in Charge of You Are Still in My Heart Team), won the University-Level Third Prize, Advanced Individual, School-Level First Prize, etc. The practice deeds were reported by various media such as China Youth Network, Civilized Jiangxi and Civilized Ganzhou.
- Served as a part-time league cadre for college students in Shangyou County, Jiangxi Province from 2022 to 2023 for one year, assisting the Communist Youth League Committee of Sixia Town in carrying out publicity and practice activities.
- Has been a member of Shangyou County Young Volunteers Association since Aug 2022, and has participated in various volunteer service activities for many times.

According to incomplete statistics, the total duration of my social practice and volunteer service has exceeded 1500 hours.

Internship Experience

KSIB-E-Commerce, Kuaishou, Overseas E-commerce Logistics Operation (Analysis), Daily Intern, Jun 2025 - Sep 2025

- Construction of logistics data monitoring system: Led the improvement of the development of logistics BI dashboard for data monitoring and drafted training materials, improving data query efficiency by about 50%; queried core data such as abnormal order trajectories and overdue order details with SQL, and made decisions on subsequent processing measures.
- Implementation of W2D model: Led the docking of W2D model for logistics providers such as J&T-Loggi and J&T-Anjun, and formulated standardized SOP for first-mile and last-mile handover; followed up on newly accessed test orders, coordinated issues of compensation confirmation and pickup efficiency, and processed more than 12,000 abnormal packages in handover.
- Logistics compliance and ROI optimization: Led the CTe invoicing transformation of new entities, collaborated with the finance, taxation and legal department, drafted communication documents with logistics providers, and coordinated multi-department participation in problem discussion; compiled programs to analyze the trend of logistics provider price card distribution, providing certain support for order allocation strategies.

Social Experience

Changchun XingJiu Technology Co., Ltd., Founder & CEO, Dec 2022 - Apr 2024

- Responsible for the overall management of the company, product design, constitution compilation, contract formulation, etc.; led the team to participate in innovation and entrepreneurship competitions, and drafted intellectual property achievements such as patents and software copyrights.

Yibin Qianxiao Lianpin Network Technology Co., Ltd., Person in Charge of Jilin University, Jan 2023 - Dec 2023

- Responsible for building the university operation team, recruiting supervisors and agents, conducting product marketing and docking with partners or customers.

Communist Youth League Committee of Shangyou County, Part-time League Cadre for College Students, Mar 2023 - Mar 2024

- Responsible for building a team to assist the local Communist Youth League organization in carrying out work during holidays, and planning, organizing and publicizing the content of social practice activities.

Self-Evaluation

- Strengths: (1) Strong learning ability, able to quickly master the use of tools or business processes; (2) Outstanding organizational and leadership ability with rich experience as a team leader; (3) Good language expression ability, good at defense reports and text writing; (4) Quick innovative thinking, good at solving problems.
- Personality Traits: Positive and optimistic, hard-working, responsible, honest and trustworthy, willing to challenge, daring to think and act, and strict with oneself.